UNIT 1
KNOW YOUR COMPUTER

Learning Objectives

- Understand how computers work using the Input-Process-Output cycle.
- Learn key advantages, like speed and multitasking.
- Recognize limitations, such as needing instructions and electricity.
- Apply concepts through practical exercises.

WHAT IS A COMPUTER?

A computer is an electronic machine which can do a variety of tasks at the same time. This is the reason why a computer is being used in different areas.

Children, do you agree that a computer can do a variety of tasks? Now, let us check what you can do with your computer:

ACTIVITY

Tick the type of work that you can do on your computer.

Play Games		Draw and colour	
Type letters		Learn about various subjects	
Watch movies		Send and receive e-mails	
Store files		Make food	

HOW DOES A COMPUTER WORK?

A computer is an electronic machine that takes data (instructions) as input, processes it, and provides the result as output. Computer follows "Input-Process-Output" cycle.

• INPUT, PROCESSING, OUTPUT

• INPUT

You enter data and instructions using input devices (e.g., keyboard or mouse). Instructions tell the computer what to do with the data.

• PROCESSING

The computer processes the input data with the help of its processing unit (e.g., CPU). CPU is the brain of computer, responsible for executing instructions and performing calculations to process data.

• OUTPUT

After processing, the computer gives the result on output devices (e.g., monitor/LCD or printer).

This cycle is known as the **Input-Process-Output (IPO) process.**

• EXAMPLE:

FLOUR
INPUT

KNEAD
PROCESS

DOUGH
OUTPUT

INSTRUCTION
ADD
DATA
5, 10

INPUT

$$\begin{array}{r} 5 \\ +10 \\ \hline 15 \\ \hline \end{array}$$

PROCESS

15

OUTPUT

ADVANTAGES OF A COMPUTER

- **WORKS FAST**

Computers perform tasks like calculations or writing faster than humans.

- **STORES INFORMATION**

Computers can store a lot of files, pictures, and videos.

- **HELPS IN LEARNING**

Computers help students learn new subjects and skills.

- **PERFORMS MULTIPLE TASKS**

Computers can perform many tasks at the same time, like playing games, typing, and browsing the internet.

- **HELPS IN COMMUNICATION**

Computers allow us to send and receive emails, making communication faster and easier.

- **ACCURATE**

Computers can perform all calculations accurately.

LIMITATIONS OF A COMPUTER

- **NEEDS INSTRUCTIONS**

A computer cannot work on its own; it needs instructions from a user.

- **NEEDS ELECTRICITY**

A computer only works when it has power.

- **CANNOT THINK**

Computers cannot make decisions or solve problems on their own without human guidance.

- **REQUIRES MAINTENANCE**

Computers need regular updates and maintenance to function properly.

- **LIMITED BY PROGRAMMING**

A computer can only do what it is programmed to do; it cannot do tasks outside its programming.

EXERCISE 1

Fill in the Blanks.

1. A computer works by following the ______________ given by the user.

2. After the data is processed, the computer gives the result on ______________ devices.

3. Computers can store a large amount of ______________.

4. The processing unit of a computer is called the ______________.

5. A computer can only work when it has ______________.

6. To enter data into the computer, we use ______________ devices like a keyboard.

EXERCISE 2

Mark "True" or "False".

	TRUE	FALSE
A computer can work without any instructions.		
A computer can process and show results on output devices like a monitor.		

Computers are used in only one field of work.		
You can draw and colour using a computer.		
A computer can work even if there is no electricity.		
Computers work faster than humans when doing tasks like calculations.		

EXERCISE 3

Answer the following questions.

1. What are the three main steps in how a computer works?

Ans: ______________________________

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2. Write two advantages of using a computer.

Ans: ______________________________

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3. Write two limitations of a computer.

Ans: ______________________________

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4. Why does a computer need electricity?

Ans: ______________________________

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6. Why do computers need instructions from a user to work?

Ans: ______________________________

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EXERCISE 4

Classify the following devices as Input, Output or Processing.

Devices	
Keyboard	
Mouse	
CPU	
Monitor/LCD	